

Admin Husic, Ph.D.

Associate Professor • Civil and Environmental Engineering • Virginia Tech
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EDUCATION

Doctor of Philosophy (Ph.D.), Civil Engineering, University of Kentucky	July 2018
Master of Science (MS), Civil Engineering, University of Kentucky	Dec 2015
Bachelor of Science (BS), Civil Engineering, University of Kentucky	May 2014

ACADEMIC APPOINTMENTS

Associate Professor Aug. 2024 – Present
 Occoquan Watershed Monitoring Laboratory
 Department of Civil and Environmental Engineering
 Virginia Tech (VT), Blacksburg, VA

Affiliate Faculty Apr. 2026 – Present
 The Global Change Center
 Virginia Tech (VT), Blacksburg, VA

Assistant Professor Aug. 2018 – Aug. 2024
 Department of Civil, Environmental, and Architectural Engineering
 University of Kansas (KU), Lawrence, KS

MENTORING EXPERIENCE

Primary Research Advisor	Total	Current
Postdoctoral Fellow	2	1
Doctoral Student	5	4
Masters Student	8	-
Undergraduate Student	14	-
External Committee Member		
Doctoral Student	11	8
Masters Student	2	-

PEER-REVIEWED JOURNAL ARTICLES (research advisees indicated by underscores)

In Review

1. **Husic, A.**, Speir, S., Tank, J., Zarnetske, J., Kurz, M., McMillan, H., Percich, A., Kemper, J., Miralha, L., Byrnes, D., Chang, S., Wellen, C., Pradhanang, S., Peng, B., Kuntla, S. K., Zhang, Q., Baldus, J., Liu, X., Bergstrom, A., Warix, S., Stewart, B., Kelly, P., Mahoney, T., Corman, J., Chen, C., Lintern, A., Castronova, T., and Garousi Nejad, I. (2026).

Harmonizing fragmented sensor networks to transform water quality science and management. *In Review*.

2. Swenson, L., Zipper, S., and **Husic, A.** (2026). Hydrological processes driving deep learning streamflow model deficiencies in flat, human-modified landscapes. *In Review*.
3. Zhang, Q., Sabo, R., Appling, A., Baker, M., Dennison, W., Golden, H., Huang, J., **Husic, A.**, Linker, L. C., Liu, Y., Maloney, K., Shen, C., Shen, J., Shen, G., Van Meter, K., Zheng, Y., and Zhi, W. (2026). Making waves: Closing the information-to-action gap for artificial intelligence in watershed and estuarine management. *In Review*.
4. Liu, F., Zarnaghsh, A., **Husic, A.**, and Li, L. (2026). Extreme storms intensified water quality responses most strongly in low-urban rivers. *In Revision*.
5. Stewart, B., Li, L., Kelleher, C., Christensen, J., Golden, H., Sadayappan, K., **Husic, A.**, Seybold, E., McMillan, H., Zimmer, M., Lane, C., Sando, R., Jones, C., Segura, C., Price, A., Jaeger, K., Cheng, F., Mahoney, D., Kondum, F., Hatley, C., Sabel, P., Hammond, J., and Ward, A. (2026). Small streams experience less persistent but more intense heatwaves than large rivers. *In Review*.
6. Liu, H., Peng, B., Ma, Z., Li, L., Speir, S. L., **Husic, A.**, Sivapalan, M., Guan, K., Pan, M., Zhao, Q., Wang, Y., Jia, M., and Yang, J. (2026). Nitrate availability and hydrologic connectivity coregulate event-scale concentration contrast and C–Q patterns. *In Review*.
7. Hale, C., Hotchkiss, R., Shelley, J., **Husic, A.**, Minear, T., and Mansfield, M. (2026). Cost-effectiveness model for reservoir sediment management economics. *In Review*.

Published

1. Percich, A., Gellis, A., Fox, J., and **Husic, A.** (2026). Upscaling sediment source prediction for watershed management. *Communications Earth & Environment*, *In Press*.
2. Sigdel, N. and **Husic, A.** (2026). U.S. rivers are transporting more suspended sediment, often in less time. *Communications Earth & Environment*, *In Press*.
3. Chen, C. and **Husic, A.** (2026). Streamflow composition in U.S. rivers is shifting toward recent precipitation. *Communications Earth & Environment*, *In Press*.
4. Appel, E., Enderami, A., Sutley, E., and **Husic, A.** (2026). Disproportionate exposure to extreme floods in historically redlined U.S. communities. *Nature Communications*, *17*, 8990.
5. Zang, Y., Treble, P., Yoshimura, K., Pinza, J., Zhang, F., Özdemir, K., Mei, X., **Husic, A.**, Gessert, A., Stroj, A., Andreo, B., Ladouche, B., Stumpp, C., Mance, D., Zagana, E., Huang, F., Sappa, G., Kunstmann, H., Brielmann, H., Zhou, H., Wu, H., Garvelmann, J., Berglund, J., Charlier, J.-B., Lange, J., Barberá Fornell, J., Pu, J., Katsanou, K., Ren, K., Toran, L., Gill, L., Filippini, M., Kralik, M., Mudarra Martínez, M., Zhao, M., Luo, M., Goldscheider, N., Lambrakis, N., De Vita, P., Xiao, Q., Yu, S., Iacurto, S., Coda, S., McCormack, T., Allocca, V., Darling, W., D'Alessandro, W., Guo, X., Hu, Y., Wang, Z., Kaminsky, E., Faimon, J., Lang, M., Pracný, P., and Hartmann, A. (2026). The WoKaS-Iso Database: Workflow for a Global Compilation of Oxygen-18 and Deuterium Records in Karst Springs and Cave Drip Water for Enhanced Understanding of Karst Systems. *Earth System Science Data*, *18*, 7199–7225.

6. Seybold, E., Hammond, J., **Husic, A.**, McMillan, H., Golden, H., Christensen, J., Segura, C., Zimmer, M., Price, A., Jaeger, K., Li, L., Mahoney, T., Cheng, F., Jones, N., Kelleher, C., Sando, R., and Lane, C., (2026). Emergent headwater types for watershed monitoring, research, and planning. *Geophysical Research Letters*, 53(17), e2026GL123244.
7. Araki, R., Holt, A., Hammond, J., **Husic, A.**, Coxon, G., and McMillan, H. (2026). Continental-scale prediction of hydrologic signatures and processes. *Hydrology and Earth System Sciences*, 30, 3647–3673.
8. Sando, R., Jaeger, K., Kelleher, C., Hammond, J., Christensen, J., Segura, C., Golden, H., Cheng, F., **Husic, A.**, Jones, N., Lane, C., Li, L., Mahoney, T., McMillan, H., Price, A., Seybold, E., Ward, A., and Zimmer, M. (2026). Streamflow and Surface-water Presence Data Availability Across the Conterminous United States (CONUS): A Review for Headwater Systems. *Hydrological Processes*, 40(5), e70572.
9. Li, J., Wang, D., Wang, G., Ma, J., Deng, Z., Song, C., Guo, L., Li, X., **Husic, A.**, and Wang, W. (2026). Cryo-FSC: A Dynamic Functional Connectivity Index Reveals Spatio-Temporal Shifts in Sediment Transport in the Permafrost-Dominated River Systems. *Water Resources Research*, 62(5), e2025WR043141.
10. Layzell A., Lloyd, C., Percich, A., Harris, T., Ketterer, M., and **Husic, A.** (2026). Event-driven shifts in river and reservoir sediment sources: Cottonwood River and John Redmond Reservoir, USA. *Earth Surface Processes and Landforms*, 51(4), e70289.
11. Brandão, A., **Husic, A.**, Almagro, A., Schwambach, D., and Oliveira, P. (2026). Explaining spatial variation and catchment characteristics thresholds in streamflow response. *Journal of Hydrology*, 674, 135492.
12. Valencia, S., Marín, D., Gómez, D., Echavarría-Porras, V., Mejía-Sepulveda, J., **Husic, A.**, Sullivan, S., Hoyos, N., Villegas, J., Harrison, L., and Green, J. (2026). Improvements and limitations of the new Climate Hazards Center Infrared Precipitation with Stations (CHIRPSv3) dataset: Insights from multiple spatio-temporal scales in Colombia. *Atmospheric Research*, 338, 108971.
13. Riddle, B., Fox, J., Ford, W., Kim, M., **Husic, A.**, Backus, J., and Pollock, E. (2026). Continuum of denitrification to end-member controls on nitrate isotopic ratios: evidence from meta-analyses and a six-year record of seasonality from Kentucky, USA. *Science of the Total Environment*, 1029, 181715.
14. Grant, S., Bhide, S., Spiesman, A., Misra, S., Rippey, M., Galik, C., Birkland, T., Schenk, T., Kaushal, S., Vikesland, P., Knocke, W., Hotchkiss, E., **Husic, A.**, Post, H., Coneway, C., Prelewicz, G., Owsenek, B., Steglitz, B., Laursen, B., and Rowles, K. (2026). Social-ecological-technological drivers of freshwater salinization in the Occoquan Reservoir, United States. *Communications Earth & Environment*, 7, 133.
15. Lane, C., D’Amico, E., Christensen, J., Golden, H., Cheng, F., Hammond, J., **Husic, A.**, Jaeger, K., Jones, N., Kelleher, C., Li, L., Mahoney, T., McMillan, H., Price, A., Sando, R., Segura, C., Seybold, E., Ward, A., and Zimmer, M. (2026). Identifying Headwater Streams across the Conterminous United States. *Ecosystems*, 29, 26.

16. Bhide, S., Grant, S., Benettin, P., Rippey, M., Monofy, A., Furst, K., Shelton, S., Kaushal, S., Misra, S., Vikesland, P., Hotchkiss, E., Spiesman, A., Prelewicz, G., Schenk, T., Post, H., Alvi, D., Steglitz, B., and **Husic, A.** (2026). Transit times link pollution sources to drinking water quality in a “One Water” system. *Water Research*, 288, 124652.
17. Shrestha, S., Hotchkiss, R., Minear, T., Shelley, J., and **Husic, A.** (2025). Sediment storage behind low-head dams: assessing potential contribution to reservoir sedimentation in Kansas, USA. *River Research and Applications*, 41, 2177-2188.
18. Pandit, A., Golden, H., Christensen, J., Lane, C. R., and **Husic, A.** (2025). Deep learning prediction and interpretation of riverine nitrate export across the Mississippi River Basin. *Water Resources Research*, 61(8), e2024WR039207.
19. **Husic, A.**, Hammond, J., Price, A., and Roundy, J. (2025). Interrogating process deficiencies in large-scale hydrologic models with interpretable machine learning. *Hydrology and Earth System Sciences*, 29(18), 4457–4472.
20. Marin, D., Grant, S., Shantanu, B., Rippey, G., Gomez-Velez, J., Brent, R., Kaushal, S., Post, H., Shelton, S., Misra, S., Hotchkiss, E., Monofy, A., Alvi, D., Schmitz, B., Curtis, S., Davis, C., Vikesland, P., and **Husic, A.** (2025). Ion Clusters Reveal the Sources, Impacts, and Drivers of Freshwater Salinization. *Environmental Science and Technology*, 59(27), 14053–14062.
21. Li, J., Wang, G., Ma, J., Huang, P., Song, C., Sun, S., Guo, L., Li, K., Wang, H., Li, D., and **Husic, A.** (2025). Structural Connectivity Shaped by Cryospheric Degradation Modulates Sediment Transport Dynamics in High Mountain Asia. *Water Resources Research*, 61(6), e2024WR038654.
22. Spellman, P., **Husic, A.**, and Hasenmueller, E. (2025). Storm-induced solute dynamics and hysteresis behavior in an eogenetic karst aquifer. *Hydrological Processes*, 39(4), e70111.
23. Pandit, A., Hogan, S., Mahoney, D., Ford, W., Fox, J., Wellen, C., and **Husic, A.** (2025). Establishing performance criteria for evaluating watershed-scale sediment and nutrient models at fine temporal scales. *Water Research*, 274, 123156.
24. Golden, H., Christensen, J., McMillan, H., Kelleher, C., Lane, C., **Husic, A.**, Li, L., Ward, A., Hammond, J., Seybold, E., Jaeger, K., Zimmer, M., Sando, R., Jones, N., Segura, C., Mahoney, T., Price, A., and Cheng, F. (2025). Advancing the Science of Headwater Streamflow for Global Water Protection. *Nature Water*, 3(1), 16–26.
25. Riddle, B., Fox, J., Ford, W., **Husic, A.**, and Pollock, E. (2025). Fourteen-year fluvial sediment record shows non-conservativeness of organic tracers: recommendations for sediment fingerprinting. *Hydrological Processes*, 39(1), e70054.
26. Bettel, L., Fox, J., **Husic, A.**, Mahoney, T., Martin, A., Zhu, J., Tobin, B., Al-Aamery, N., Osborne, C., Riddle, B., and Pollock, E. (2025). Hydrologic pathways and baseflow contributions, and not the proximity of sediment sources, determine the shape of sediment hysteresis curves: Theory development and application in a karst basin in Kentucky USA. *Journal of Hydrology*, 649, 132300.
27. Clune, J., Cravotta, C., **Husic, A.**, Abraham, H., and Schmidt, K. (2024). Complex hydrology and variability of nitrogen sources in a karst watershed. *Journal of Environmental Quality*, 53, 492-507.

28. Zarnaghsh, A., Kelly, M., Burgin, A., and **Husic, A.** (2024). Revealing nitrate uptake and dispersion dynamics using high-frequency sensors and two-dimensional modeling in a large river system. *Advances in Water Resources*, 187, 104693.
29. Rider, Z., Percich, A., Hiripitiyage, Y., Harris, T., Sturm, B., Wilson, A., Pollock, E., Beaver, J., and **Husic, A.** (2024). Drivers of cyanotoxin and taste-and-odor compound presence within the benthic algae of human-disturbed rivers. *Water Research*, 253, 121357.
30. Michalek, A., Villarini, G., and **Husic, A.** (2023). Climate change projected to impact structural hillslope connectivity at the global scale. *Nature Communications*, 14, 6788.
31. Zarnaghsh, A. and **Husic, A.** (2023). An index for inferring dominant transport pathways of solutes and sediment: assessing land use change with high-frequency conductivity and turbidity sensor data. *Science of the Total Environment*, 894, 164931.
32. McVey, Z., Michalek, A., Mahoney, T., and **Husic, A.** (2023). Urbanization as a limiter and catalyst of watershed-scale sediment transport: insights from probabilistic connectivity modeling. *Science of the Total Environment*, 894, 165093.
33. **Husic, A.**, Fox, J., Clare, E., Mahoney, T., and Zarnaghsh, A. (2023). Nitrate hysteresis as a tool for revealing storm-event dynamics and improving water quality model performance. *Water Resources Research*, 59(1), e2022WR033180.
34. Gerlitz, M., Fox, J., Ford, W., **Husic, A.**, Mahoney, T., Armstead, M., Hendricks, S., Crain, A., Backus, J., Pollock, E., Wei, R., Bo, T., Riddle, B., and White, D. (2023). Nitrate sensors suggest soil-plant processes produce three distinct seasonal patterns of nitrate in the Ohio River Basin. *Journal of the American Water Resources Association*, 59(4), 635-651.
35. **Husic, A.** and Michalek, A. (2022). Structural hillslope connectivity is driven by tectonics more than climate and modulates hydrologic extremes and benefits. *Geophysical Research Letters*, 49(15), e2022GL099898. *Selected by Editors as a Research Spotlight.
36. Percich, A., **Husic, A.**, and Ketterer, M. (2022). Plutonium isotopes: an effective tool for fluvial sediment sourcing in urbanized catchments. *Geophysical Research Letters*, 49(2), e2021GL094497.
37. **Husic, A.**, Al Aamery, N., and Fox, J. (2022). Simulating hydrologic pathway contributions in fluvial and karst settings: an evaluation of conceptual, physically-based, and deep learning modeling approaches. *Journal of Hydrology X*, 17, 100134.
38. Bettel, L., Fox, J., **Husic, A.**, Zhu, J., Al-Aamery, N., Mahoney, T., and Gold-McCoy, A. (2022). Sediment transport investigation in a karst aquifer hypothesizes controls on internal versus external sediment origin and saturation impact on hysteresis. *Journal of Hydrology*, 613, 128391.
39. Michalek, A., **Husic, A.**, Roundy, J., and Hansen, A. (2021). Assessment of climatic and anthropogenic controls on bridge deck drainage and sediment removal. *Water*, 13(24), 3556.
40. Zarnaghsh, A. and **Husic, A.** (2021). Degree of anthropogenic land disturbance controls fluvial sediment hysteresis. *Environmental Science and Technology*, 55(20), 13737–13748.
41. Kelly, M., Zeglin, L., **Husic, A.**, and Burgin, A. (2021). High Supply, High Demand: A Unique Nutrient Addition Decouples Nitrate Uptake and Metabolism in a Large River. *Journal of Geophysical Research: Biogeosciences*, 126(12), e2021JG006469.

42. **Husic, A.**, Fox, J., Al-Aamery, N., Pollock, E., Ford, W., Agouridis, C., and Backus, J. (2021). Seasonality of recharge drives spatiotemporal nitrate removal in a karst conduit as evidenced by sediment nitrogen isotopes. *Journal of Geophysical Research: Biogeosciences*, 126(10), e2021JG006454.
43. Michalek, A., Zarnaghsh, A., and **Husic, A.** (2021). Modeling linkages between erosion and connectivity in an urbanizing landscape. *Science of the Total Environment*, 764, 144255.
44. Al Aamery, N., Adams, E., Fox, J., **Husic, A.**, Zhu, J., Gerlitz, M., Agouridis, C., and Bettel, L. (2021). Numerical model development for investigating hydrologic pathways in shallow fluviokarst. *Journal of Hydrology*, 593, 125844.
45. **Husic, A.**, Fox, J., Mahoney, T., Gerlitz, M., Pollock, E., and Backus, J. (2020). Optimal Transport for Assessing Nitrate Source-Pathway Connectivity. *Water Resources Research*, 56(10), e2020WR027446.
46. **Husic, A.**, Fox, J., Adams, E., Pollock, E., Ford, W., Agouridis, C., and Backus, J. (2020). Quantification of nitrate removal in karst conduits using stable isotopes and numerical modeling. *Water Research* (170), 115348.
47. Ford, W., **Husic, A.**, Fogle, A., and Taraba, J. (2019) Long-term assessment of nutrient flow pathway dynamics and in-stream fate in a temperate karst agroecosystem watershed. *Hydrological Processes*, 33 (11), 1610-1628.
48. **Husic, A.**, Fox, J., Ford, W., Agouridis, C., Currens, J., and Backus, J. (2019). Nitrate pathways, processes, and timing in an agricultural karst system: development and application of a numerical model. *Water Resources Research*, 55(3), 2079-2103.
49. Ghesemi, A., Fox, J., and **Husic, A.** (2019). Predicting macroturbulence energy and timescales for open channel flow over a gravel bed: experimental results and scaling laws. *Geomorphology*, 322, 122-137.
50. **Husic, A.**, Fox, J., Adams, E., Backus, J., Pollock, E., Ford, W., and Agouridis, C. (2019) Inland impacts of atmospheric river and tropical cyclone extremes on nitrate transport and stable isotope measurements. *Environmental Earth Science*, 78(36).
51. Jensen, A., Ford, W., Fox, J., and **Husic, A.** (2018). Improving water quality models using stable isotope tracers: a review and synthesis. *Transactions of the American Society of Agricultural and Biological Engineers*, 61, 139-157.
52. **Husic, A.**, Fox, J., Agouridis, C., Currens, J., Ford, W., and Taylor, C. (2017). Sediment carbon source, fate, and transport in a fluviokarst watershed (Part 1): conceptual model development. *Journal of Hydrology*, 549, 179-193.
53. **Husic, A.**, Fox, J., Ford, W., Agouridis, C., Currens, J., and Taylor, C. (2017). Sediment carbon source, fate, and transport in a fluviokarst watershed (Part 2): numerical model development and application. *Journal of Hydrology*, 549, 208-219.

AWARDED GRANTS as PI (by date) (total: \$3,036,069 since 2019)

1. **Husic, A.**, Post, H., and Alvi., D. 2025–2027. The Broad Run Watershed Monitoring Program in Loudoun County, Virginia. Loudoun Water. **\$402,744.**

2. **Husic, A.**, Post, H., and Alvi., D. 2025–2027. Broad Run PFAS Sampling. Loudoun Water. **\$103,773.**
3. **Husic, A.** 2024–2029. CAREER: Dynamic connectivity: a research and educational frontier for sustainable environmental management under climate and land use uncertainty. National Science Foundation. **\$609,734.**
4. **Husic, A.** and Layzell, T. 2024–2026. A statewide inventory of the dominant reservoir sedimentation sources to inform targeted watershed conservation and upstream mitigation practices. Kansas Water Office. **\$210,387.**
5. **Husic, A.**, Gido, K., Harris, T., and Bigham, K. 2023–2026. Collaborative Research: Can Human-Induced Turbidity Currents Enable Sustainability of Freshwater Reservoirs? National Science Foundation. **\$500,000** (\$350,000 to KU, \$150,000 to KSU).
6. **Husic, A.** 2023–2027. Collaborative Research: GCR: Common Pool Resource Theory as a Scalable Framework for Catalyzing Stakeholder-Driven Solutions to the Freshwater Salinization Syndrome. National Science Foundation. (Former PI: Jesus Gomez-Velez). **\$293,843.**
7. **Husic, A.** 2023–2024. RII Track-4: Assessing Dynamic Connectivity of Streams and Wetlands across Spatial and Human Gradients with Deep Learning. National Science Foundation. **\$259,515.**
8. **Husic, A.** 2023–2024. The inequity of urban flooding: exploring societal and climatic drivers across scales. NSF Kansas EPSCoR ARISE Program. **\$50,000.**
9. **Husic, A.** 2022–2024. Monitoring and modeling Hillsdale Lake under changing land use and climate drivers to assess tipping point of hypereutrophic lake conditions. Kansas Department of Health and the Environment. **\$92,142.**
10. **Husic, A.** and Zipper, S. 2022–2024. Forecasting streamflow and groundwater depletion with deep learning models to sustain Kansan water resources. Kansas Water Resources Research 104(b) Program. **\$50,000** (+\$50,000 in-kind cost share).
11. **Husic, A.** 2022–2023. How does urbanization change nitrate source-zone activation? Real-time insights from rapidly-expanding Johnson County, Kansas. University of Kansas Office of Research General Research Fund. **\$13,000.**
12. **Husic, A.** and Sutley, E. 2022–2023. Evaluating the Racial Equity of Flooding Hazards in the Kansas City Metropolitan Area. KU Racial Equity Research, Scholarship & Creative Activity Fund. **\$19,607.**
13. **Husic, A.** 2021–2024. Identifying the Source of Sediment for Targeted Best Management Practices in Urban and Rural Areas of Johnson County. Johnson County Stormwater Management Program. **\$175,175.**
14. **Husic, A.** 2021–2022. Influences of urbanization on the tipping points of benthic algae mat growth, decay, and toxin production. NSF Kansas EPSCoR MAPS Program. **\$141,777.**
15. **Husic, A.**, Collins, W., Lequesne, R., and Roundy, J. 2021–2022. Development of Condition Factors and Bridge Load Ratings through Statistical Analyses of the NBI Database. Kansas Department of Transportation. **\$46,500.**

16. **Husic, A.**, Harris, T., Sturm, B., 2020–2022. Benthic cyanobacterial mats: a potential source of harmful and nuisance compounds to Kansas streams. Kansas Water Resources Research 104(b) Program. **\$40,000** (+\$80,000 in-kind cost share).
17. **Husic, A.** 2020–2021. Sediment tracing with plutonium isotopes to assess urbanization impact on sources of erosion. University of Kansas Office of Research General Research Fund. **\$7,872**.
18. **Husic, A.** 2019–2021. Optimal Transport for Assessing Nitrate Source and Pathway Connectivity in Human-Disturbed Watersheds. University of Kansas Office of Research New Faculty General Research Fund. **\$20,000**.

AWARDED GRANTS as Co-PI (by date) (total: \$8,222,877 since 2019)

1. Hotchkiss, R., **Husic, A.**, and Minear, T. 2023–2025. Understanding Reservoir Sedimentation Sources and Economics in the Kansas River Basin for US Army Corps of Engineers Kansas City District. U.S. Army Corps of Engineers. **\$316,000** (\$100,000 to KU/VT, \$166,000 to BYU, and \$50,000 to CUB).
2. Bennet, C., Lequesne, R., Li, J., Collins, W., Lepage, A., Darwin, D., O'Reilly, M., Hansen, A., and **Husic, A.** 2022–2027. Improving the Performance of Concrete Dam Infrastructure Through Use of Fiber-Reinforced Polymers. U.S. Army Corps of Engineers. **\$7,740,921**.
3. Layzell, T. and **Husic, A.** 2023–2024. Streambank Evaluation of the Cottonwood and Neosho Rivers above John Redmond Reservoir. Kansas Water Office. **\$100,000**.
4. Roundy, J., Hansen, A., and **Husic, A.** 2018–2020. Bridge Deck Drainage: Evaluation of KDOT's Current Design Guidance. Kansas Department of Transportation. **\$65,956**.

INVITED RESEARCH PRESENTATIONS (by me as presenting author)

1. **Husic, A.** (2026, December). Lessons learned from large-sample machine learning of river water quality. American Geophysical Union Fall Meeting. San Francisco, CA. Oral Presentation.
2. **Husic, A.** (2026, November). What big data can do for water quality: reconstructing and attributing change in rivers. University of Florida. Water, Wetlands, and Watersheds Seminar Series. Gainesville, FL. Virtual Oral Presentation.
3. **Husic, A.** (2026, November). Harmonized aquatic big data and deep learning for catchment-scale inference. University of Connecticut. Department of Earth Sciences Seminar. Storrs, CT. Virtual Oral Presentation.
4. **Husic, A.** (2026, July). Harmonizing fragmented sensor networks to transform water quality science and management. United States Geological Survey. Water Resources Mission Area Seminar Series. Reston, VA. Virtual Oral Presentation.

5. **Husic, A.**, and Chen, C. (2026, May). American rivers are shifting toward younger streamflow. North Dakota State University. North Dakota Water Resources Research Institute Seminar Series. Fargo, ND. Virtual Oral Presentation.
6. **Husic, A.** (2025, December). From short bursts to long trends: disentangling human and climate influence on sediment connectivity across time and space. American Geophysical Union Fall Meeting. New Orleans, LA. Oral Presentation.
7. **Husic, A.** (2025, October). American rivers are transporting more sediment in less time. University of Maryland. Geology Colloquium. College Park, MD. Oral Presentation.
8. **Husic, A.** (2025, June). Learning from landscapes: deep learning and interpretable machine learning to advance catchment science. Gordon Research Conference on Catchment Science. Andover, NH. Oral Presentation.
9. **Husic, A.** (2024, December). Built environments modulate sediment delivery: results from local to continental scales. American Geophysical Union Fall Meeting. Washington, DC. Oral Presentation.
10. **Husic, A.** (2024, December). Assessing continental gradients in the significance of runoff and baseflow to turbidity generation in streams. University of Maryland – Baltimore County. Center for Urban Environmental Research and Education. Baltimore, MD. Oral Presentation.
11. **Husic, A.** (2023, December). Urban connectivity: pathways and processes of solute and sediment delivery to rivers. American Geophysical Union Fall Meeting. San Francisco, CA. Oral Presentation.
12. **Husic, A.** (2023, February). Using radioactive fallout to quantify streambank erosion in urbanizing Johnson County, Kansas. Mid-America Regional Stormwater Council. KC Urban Stormwater Conference. Lenexa, KS. Oral Presentation.
13. **Husic, A.** (2022, December). Hydrologic connectivity: a management frontier for water quality conservation. American Geophysical Union Fall Meeting. Chicago, IL. Oral Presentation.
14. **Husic, A.** (2022, November). Hydrologic connectivity in the Anthropocene: sensing and untangling pathways of water and sediment from local to continental scales. Lawrence Berkely National Lab. Environmental and Earth Sciences Modeling Forum. Berkeley, CA. Oral Presentation.
15. **Husic, A.** (2022, November). Nitrogen fate and transport in fluvio-karst systems: evidence from high-frequency sensing, stable isotope tracing, and numerical modeling. Chesapeake Bay Factors Team Meeting. Virtual Oral Presentation.
16. **Husic, A.** (2021, March). Untangling the Controls of Water and Nutrient Connectivity in Karst Aquifers. Kansas State University Geology Seminar. Manhattan, KS. Virtual Oral Presentation.

STUDENT PRESENTATIONS (by student or postdoc advisee, * indicates award)

1. Chen, C. and **Husic, A.** (2026, June). Increasing event water fraction across the Chesapeake Bay Watershed under climatic and anthropogenic change. Chesapeake Community Research Symposium. Annapolis, MD. Oral Presentation.
2. Percich, A., Gellis, A., Fox, J., and **Husic, A.** (2026, June). Watershed-scale sediment source prediction using machine learning. Chesapeake Community Research Symposium. Annapolis, MD. Oral Presentation.
3. Marin, D., Chen, C., Grant, S., Kaushal, S., and **Husic, A.** (2026, June). Capturing event-driven salinity pulses and nonlinear SC dynamics in Chesapeake Bay tributaries using a deep learning model. Chesapeake Community Research Symposium. Annapolis, MD. Oral Presentation.
4. Chen, C. and **Husic, A.** (2025, December). Event water fraction increases across the conterminous United States under climatic and anthropogenic change. American Geophysical Union Fall Meeting. New Orleans, LA. Oral Presentation.
5. Percich, A. and **Husic, A.** (2025, December). Sediment fingerprinting for unmonitored basins using machine learning. American Geophysical Union Fall Meeting. New Orleans, LA. Oral Presentation.
6. Marin, D., Chen, C., Grant, S., and **Husic, A.** (2025, December). Detecting long-term freshwater salinization trends and chronic chloride exceedance with a deep learning model. American Geophysical Union Fall Meeting. New Orleans, LA. Oral Presentation.
7. Shrestha, S. and **Husic, A.** (2025, December). Why are some watersheds more sediment-productive than others? American Geophysical Union Fall Meeting. New Orleans, LA. Oral Presentation.
8. Sigdel, N. and **Husic, A.** (2025, December). Deep learning reveals the bursty and seasonal nature of continental-scale sediment transport. American Geophysical Union Fall Meeting. New Orleans, LA. Oral Presentation.
9. Lloyd, C., Layzell, A., Harris, T., Ketterer, M., and **Husic, A.** (2025, December). Sediment fingerprinting reveals event-driven shifts in sediment sources for a large federal reservoir, Kansas, USA. American Geophysical Union Fall Meeting. New Orleans, LA. Poster Presentation.
10. Kuntla, S., Speir, S., and **Husic, A.** (2025, December). STREAM: a comprehensive high-frequency sensor database of water quality in rivers in the United States. American Geophysical Union Fall Meeting. New Orleans, LA. Poster Presentation.
11. Lloyd, C., Layzell, A., Harris, T., Ketterer, M., and **Husic, A.** (2025, November). Sediment fingerprinting reveals event-driven shifts in sediment sources for a large federal reservoir, Kansas, USA. American Geophysical Union Fall Meeting. New Orleans, LA. Poster Presentation.
12. Lloyd, C.*, Layzell, A., Percich, A., Harris, T., Ketterer, M., and **Husic, A.** (2025, November). Event-driven shifts in river and reservoir sediment sources: Cottonwood River and John Redmond Reservoir, USA. Governor's Conference on the Future of Water in Kansas. Manhattan, KS. Poster Presentation.

**Awarded First Place Outstanding Graduate Poster Presentation.*

13. Chen, C. and **Husic, A.** (2025, June). Deep learning for hydrograph separation across the contiguous United States. Gordon Research Conference on Catchment Science. Andover, NH. Poster Presentation.
14. Marin, D., Chen, C., Grant, S., and **Husic, A.** (2025, June). A continental-scale long short-term memory model for predicting freshwater salinization. Gordon Research Conference on Catchment Science. Andover, NH. Poster Presentation.
15. Chen, C. and **Husic, A.** (2025, May). Predicting runoff and baseflow hydrograph components at a continental scale with deep learning. HydroML Symposium. Irvine, CA. Oral Presentation.
16. Sigdel, N. and **Husic, A.** (2025, May). Suspended sediment prediction with long short-term memory models. HydroML Symposium. Irvine, CA. Oral Presentation.
17. Marin, D., Chen, C., Grant, S., and **Husic, A.** (2025, May). A continental-scale long short-term memory model for predicting freshwater salinization. HydroML Symposium. Irvine, CA. Poster Presentation.
18. Cappotto, K., Harris, T., Gido, K., Moore, T., and **Husic, A.** (2025, May). Novel Dredging in Tuttle Creek Lake – Physical Properties & Effects on Water Quality. University of Kansas Environmental Engineering Conference. Lawrence, KS. Poster Presentation.
19. Shrestha, S., Hotchkiss, R., Minear, T., and **Husic, A.** (2024, December). Evaluating the potential for low-head dams to serve as major sediment sources to federal reservoirs in Kansas. American Geophysical Union Fall Meeting. Washington, DC. Poster Presentation.
20. Sigdel, N. and **Husic, A.** (2024, December). Integrating sensor data and deep learning for CONUS-scale suspended sediment prediction. American Geophysical Union Fall Meeting. Washington, DC. Poster Presentation.
21. Percich, A. and **Husic, A.** (2024, December). Identifying erosional sources in an urbanizing landscape using sediment fingerprinting. American Geophysical Union Fall Meeting. Washington, DC. Poster Presentation.
22. Cappotto, K., Harris, T., Gido, K., Moore, T., and **Husic, A.** (2024, November). Monitoring Reservoir Conditions in Tuttle Creek Lake Prior to Water Injection Dredging Demonstration. Governor's Conference on the Future of Water in Kansas. Manhattan, KS. Poster Presentation.
23. Pandit, A. and Golden, H., Christensen, J., Lane, C., **Husic, A.** (2024, June). Transfer learning for the prediction of nitrate loads in ungauged basins. WaterSciCon. St. Paul, MN. Oral Presentation.
24. Percich, A., VanderMeulen, L., and **Husic, A.** (2024, June). Sediment source prediction for ungauged basins with machine learning. WaterSciCon. St. Paul, MN. Oral Presentation.
25. Pandit, A. and Golden, H., Christensen, J., Lane, C., **Husic, A.** (2023, December). A continental-scale long short-term memory model for predicting nitrate dynamics. American Geophysical Union Fall Meeting. San Francisco, CA. Oral Presentation.
26. Mohn, A. and **Husic, A.** (2023, December). Microplastics in Stormwater: Quantifying Abundance, Characterizing Sources, and Understanding Associations with Water

- Parameters. American Geophysical Union Fall Meeting, San Francisco, CA. Poster Presentation.
27. Appel, E. Enderami, A., Sutley, E., and **Husic, A.** (2023, December). Evaluating the racial inequity of flooding: historical policies, modern consequences. American Geophysical Union Fall Meeting, San Francisco, CA. Poster Presentation.
 28. Pandit, A. and **Husic, A.** (2023, November). Forecasting the tipping points of hypereutrophic lake conditions under changing climate and land use scenarios. Governor's Conference on the Future of Water in Kansas. Manhattan, KS. Poster Presentation.
 29. Mohn, A.* and **Husic, A.** (2023, November). Microplastics within Fluvial Sediment: Quantifying Abundance, Characterizing Sources, and Understanding Drivers. Governor's Conference on the Future of Water in Kansas. Manhattan, KS. Poster Presentation.
**Awarded First Place Outstanding Undergraduate Poster Presentation.*
 30. Appel, E. Enderami, A., Sutley, E., and **Husic, A.** (2023, November). Troubled waters: redlined communities and present-day flooding. Governor's Conference on the Future of Water in Kansas. Manhattan, KS. Poster Presentation.
 31. Percich, A. and **Husic, A.** (2023, November). Sediment Fingerprinting to Assess Sources of Erosion in Urbanizing Johnson County. Governor's Conference on the Future of Water in Kansas. Manhattan, KS. Poster Presentation.
 32. Cappotto, K., Harris, T., Gido, K., Moore, T., and **Husic, A.** (2023, November). Water injection dredging and human-induced turbidity currents as an alternative strategy for reservoir sedimentation management. Governor's Conference on the Future of Water in Kansas. Manhattan, KS. Poster Presentation.
 33. Zarnaghsh, A. and **Husic, A.** (2023, May). Assessing continental gradients in the significance of runoff and baseflow to turbidity generation in streams. SEDHYD - Sedimentation and Hydrologic Modeling Conference. St. Louis, MO. Oral Presentation.
 34. Pandit, A. and **Husic, A.** (2023, May). Get with the timeseries: establishing metrics for water quality model performance. HydroML Symposium. Berkeley, CA. Poster Presentation.
 35. Zarnaghsh, A. and **Husic, A.** (2022, December). Coupling High-frequency Sensing and 2-d Numerical Modelling to Analyze Nitrate Dynamics in a Large River. American Geophysical Union. Fall Meeting. Chicago, IL. Oral Presentation.
 36. McVey, Z., **Husic, A.**, Michalek, Z., and Mahoney, T. (2022, December). Land Use Change Impacts on Dynamic Sediment Connectivity in an Urbanizing Region. American Geophysical Union. Fall Meeting. Chicago, IL. Poster Presentation.
 37. Pandit, A. and **Husic, A.** (2022, November). Forecasting the Tipping Points of Hypereutrophic Lake Conditions Under Changing Climate and Land Use Scenarios. Governor's Conference on the Future of Water in Kansas. Manhattan, KS. Poster Presentation.
 38. Zarnaghsh, A. and **Husic, A.** (2022, November). Quantifying Runoff and Baseflow Controls on Stream Turbidity. Governor's Conference on the Future of Water in Kansas. Manhattan, KS. Poster Presentation.

39. Vaez, M. and **Husic, A.** (2022, November). Sediment fingerprinting as a tool for identifying sources of erosion in Johnson County, Kansas. Governor's Conference on the Future of Water in Kansas. Manhattan, KS. Poster Presentation.
40. Rider, Z., **Husic, A.**, Sturm B., and Harris, T. (2022, November). Widespread presence of cyanotoxins and taste-and-odor compounds within benthic algae of human-disturbed rivers. Governor's Conference on the Future of Water in Kansas. Manhattan, KS. Poster Presentation.
41. McVey, Z., **Husic, A.**, Michalek, Z., and Mahoney, T. (2022, November). Land Use Impacts on Dynamic Sediment Connectivity Across an Urbanizing Region. Governor's Conference on the Future of Water in Kansas. Manhattan, KS. Poster Presentation.
42. Rider, Z., **Husic, A.**, Sturm B., and Harris, T. (2022, April). Algae mats as sources of cyanotoxins and taste-and-odor compounds. University of Kansas Environmental and Water Resources Engineering Seminar Series. Lawrence, KS. Poster Presentation.
43. Zarnaghsh, A. and **Husic, A.** (2021, December). High-Frequency Sensing of Specific Conductance to Understand Transport Mechanisms Across an Urbanization Gradient. American Geophysical Union. New Orleans, LA. Poster Presentation.
44. Rider, Z., **Husic, A.**, Sturm B., and Harris, T. (2021, December). Land Use Impacts on the Development of Cyanotoxins and Taste-and-Odor Compounds in Benthic Algae Mats. American Geophysical Union. New Orleans, LA. Poster Presentation.
45. McVey, Z. and **Husic, A.** (2021, December). Land Use Impacts on Dynamic Sediment Connectivity Across an Urbanizing Region. American Geophysical Union. New Orleans, LA. Poster Presentation.
46. Zarnaghsh, A.*. and **Husic, A.** (2021, November). High-Frequency Sensing of Specific Conductance to Understand Solute Transport Mechanisms During Storm Events in Johnson County, KS. Governor's Conference on the Future of Water in Kansas. Virtual. Poster Presentation.
**Awarded First Place Outstanding Graduate Poster Presentation.*
47. Rider, Z., **Husic, A.**, Sturm B., and Harris, T. (2021, November). Investigating the drivers of cyanotoxins and taste-and-odor compounds in streambed algal mats. Governor's Conference on the Future of Water in Kansas. Virtual. Poster Presentation.
48. McVey, Z. and **Husic, A.** (2021, November). Modeling streamflow in rural and urban basins: applications to Johnson County, KS. Governor's Conference on the Future of Water in Kansas. Virtual. Poster Presentation.
49. Zarnaghsh, A., Kelly, M., Burgin, A., and **Husic, A.** (2021, May). Coupling high-frequency sensing and 2-D numerical modeling to analyze nitrate dynamics in a large river. Society of Freshwater Sciences. Virtual. Oral Presentation.
50. Zarnaghsh, A. and **Husic, A.** (2020, December). High-frequency sensing of sediment hysteresis to understand storm event dynamics across an urbanization gradient. American Geophysical Union Fall Meeting. Oral Presentation.
51. Michalek, A., Zarnaghsh, A., and **Husic, A.** (2020, December). Modeling linkages between erosion and connectivity in an urbanizing landscape: a coupled sensing, geomorphometric,

- and field assessment approach. American Geophysical Union Fall Meeting. Virtual. Poster Presentation.
52. Percich, A. and **Husic, A.** (2020, December). Sediment source assessment along an urban to rural transition using plutonium isotopes and sediment fingerprint modeling. American Geophysical Union Fall Meeting. Virtual. Poster Presentation.
 53. Zarnaghsh, A.*. and **Husic, A.** (2020, November). High-Frequency Sensing of Sediment Hysteresis to Understand Storm Event Dynamics in Johnson County, KS. Governor's Conference on the Future of Water in Kansas. Wichita, KS. Poster Presentation.
**Awarded First Place Outstanding Graduate Poster Presentation.*
 54. Michalek, A., Zarnaghsh, A., and **Husic, A.** (2020, November). Identifying erosion and connectivity hotspots in urbanizing Johnson County, KS. Governor's Conference on the Future of Water in Kansas. Poster Presentation. Wichita, KS. Poster Presentation.
 55. Percich, A.*. and **Husic, A.** (2020, November). Plutonium Isotopes as Tracers of Sediment in an Urbanizing Landscape. Governor's Conference on the Future of Water in Kansas. Wichita, KS. Poster Presentation.
**Awarded First Place Outstanding Undergraduate Poster Presentation.*
 56. Zarnaghsh, A. and **Husic, A.** (2019, November). High-resolution sensors for assessing the impact of rapid land use change on the processes governing stream biogeochemistry. Governor's Conference on the Future of Water in Kansas. Wichita, KS. Poster Presentation.
 57. Wilson, L. and **Husic, A.** (2019, November). Hydrologic modeling along an urbanization gradient: application of the SWAT model to five Johnson County watersheds. Governor's Conference on the Future of Water in Kansas. Wichita, KS. Poster Presentation.
 58. Percich, A.*. and **Husic, A.** (2019, November). Sediment source assessment along an urban to rural transition using stable isotopes and geochemical tracers. Governor's Conference on the Future of Water in Kansas. Wichita, KS. Poster Presentation.
**Awarded First Place Outstanding Undergraduate Poster Presentation.*
 59. Depew, A.* and **Husic, A.** (2019, November). Designing and implementing a platform for remotely sensing high-frequency water quality data across an urbanization gradient. Governor's Conference on the Future of Water in Kansas. Wichita, KS. Poster Presentation.
**Awarded Runner-up Outstanding Undergraduate Poster Presentation.*

RESEARCH PRESENTATIONS, NON-INVITED (by **me** as presenting author)

1. **Husic, A.** (2024, December). Game theory for catchment science. American Geophysical Union. Fall Meeting. Washington, DC.
2. **Husic, A.** (2024, May). Game theory for catchment science. HydroML Symposium. Richland, WA. Oral Presentation.
3. **Husic, A.** and Pandit, A. (2023, December). Large-sample water quality: exploring, modeling, and testing benchmark datasets for transferrable insights. American Geophysical Union. Fall Meeting. San Francisco, CA.

4. **Husic, A.** and Juracek, K. (2023, November). Sediment Fingerprinting of Sources of Reservoir Infilling in an Eastern Kansas Basin, USA. Oral Presentation. Governor's Conference on the Future of Water in Kansas. Manhattan, KS.
5. **Husic, A.** et al., (2023, June). Hydrologic connectivity and its influence on the pathways, processes, and timing of water quality degradation in the Anthropocene. Poster Presentation. Gordon Research Conference. Andover, NH.
6. **Husic, A.** et al., (2022, December). Benchmarking deep learning model estimates of hydrologic pathway contribution to streams and springs using conceptual and physically-based models. Poster Presentation. American Geophysical Union. Fall Meeting. Chicago, IL.
7. **Husic, A.** and Michalek, A. (2021, December). Structural hillslope connectivity modulates hydrologic extremes and benefits. Oral Presentation. American Geophysical Union. Fall Meeting. New Orleans, LA.
8. **Husic, A.** and Percich, A. (2021, November). Using plutonium fallout to quantify streambank erosion in urbanizing Johnson County, Kansas. Oral Presentation. Governor's Conference on the Future of Water in Kansas. Virtual.
9. **Husic, A.** and Michalek, A. (2021, May). Quantifying Hydrologic Connectivity at the Continental Scale: Drivers and Implications. Oral Presentation. Society of Freshwater Sciences. Virtual.
10. **Husic, A.** et al., (2020, December). High-frequency sensing of nitrate to improve numerical model performance: insights from an extensively modeled spring. Poster Presentation. American Geophysical Union Fall Meeting. San Francisco, CA.
11. **Husic, A.** et al., (2020, July). Beyond EMMA: Optimal Transport for Assessing Nitrate Source-Pathway Connectivity. Oral Presentation. Aquatic OpenScience DataScience Summit. Held virtually on Zoom & YouTube.
12. **Husic, A.** et al., (2019, December). Optimal Transport for Assessing Nitrate Source and Pathway Connectivity in a Human-Disturbed Watershed. Poster Presentation. American Geophysical Union Fall Meeting. San Francisco, CA.
13. **Husic, A.** et al., (2019, June). Sediment nitrogen stable isotopes and numerical modeling show hot moments, hot spots, and environmental drivers in the surficial fine-grained laminae of karst beds. Oral Presentation. ASCE World Environmental and Water Resources Congress. Pittsburgh, PA.
14. **Husic, A.** (2019, April). Nitrogen cycling in subsurface rivers: evidence from hydrogeochemical and isotopic modeling. Oral Presentation. Kansas Biological Survey Ecology Seminar. Lawrence, KS.
15. **Husic, A.** et al., (2018, December). Nitrate Removal in a Phreatic Karst Conduit: Estimating Nitrification and Denitrification Rates using Stable Isotopes and Numerical Modeling. Poster Presentation. American Geophysical Union Fall Meeting. Washington, D.C.
16. **Husic, A.** (2018, September). Contaminant Fate and Transport in a Karst Watershed (What Goes Down Must Come Up). Oral Presentation. University of Kansas Department of Geology Colloquium. Lawrence, KS.

17. **Husic, A.** et al., (2018, June). Application of isotopic and elemental data streams for estimating source and transformation processes in an agricultural karst system. Oral Presentation. ASCE World Environmental and Water Resources Congress. Minneapolis, MN.
18. **Husic, A.** et al., (2018, March). Water, Sediment, and Nutrient Data Streams in a Fluviokarst Watershed in the Kentucky Bluegrass: Insights from Elemental, Isotopic, and High-Resolution Sensor Data. Oral Presentation. Kentucky Water Resources Research Institute Symposium. Lexington, KY.
19. **Husic, A.** et al., (2017, December). Nitrogen fate in a phreatic fluviokarst watershed: a stable isotope, sediment tracing, and numerical modeling approach. Poster Presentation. American Geophysical Union Fall Meeting. New Orleans, LA.
20. **Husic, A.** (2017, December). Contaminant Fate and Transport in a Highly-Coupled Watershed: Integrating Stable Isotopes, Sediment Tracing, and Numerical Modeling to Assess Stream Function and Ecosystem Health. Oral Presentation. University of Kansas Department of Civil, Environmental, and Architectural Engineering Seminar. Lawrence, KS.
21. **Husic, A.** et al., (2017, May). Nitrate leaching and pathways in agricultural fluviokarst systems. Oral Presentation. ASCE World Environmental & Water Resources Congress. Sacramento, CA.
22. **Husic, A.** et al., (2017, March). Nitrate leaching in an agricultural fluviokarst system in the Bluegrass Region. Oral Presentation. Kentucky Water Resources Research Institute Symposium. Lexington, KY.
23. **Husic, A.** et al., (2016, March). Data and model investigation of a fluviokarst system in the Bluegrass Region: water, sediment, and carbon interactions. Oral Presentation. Kentucky Water Resources Research Institute Symposium. Lexington, KY.
24. **Husic, A.** et al., (2015, December). Sediment organic carbon fate and transport in a fluviokarst watershed in the Bluegrass Region. Poster Presentation. American Geophysical Union Fall Meeting. San Francisco, CA.
25. **Husic, A.** et al., (2015, December). Sediment carbon fate in a fluviokarst watershed in the Bluegrass Region. Poster Presentation. 5th Annual Sustainability Forum. Lexington, KY.
26. **Husic, A.** et al., (2015, May). Investigation of source, fate, and transport of sediments in a karst dominated watershed. Oral Presentation. ASCE World Environmental & Water Resources Congress. Austin, TX.
27. **Husic, A.** et al., (2015, March). Sediment transport mechanisms in a fluvial karst system in Bluegrass Region. Oral Presentation. Kentucky Water Resources Research Institute Annual Symposium. Lexington, KY.
28. **Husic, A.** et al., (2015, February). Sediment organic carbon fate and transport mechanisms in a fluvial karst system in the Bluegrass Region. Poster Presentation. Posters-at-the-Capitol. Frankfort, KY.

TEACHING EXPERIENCE

Courses Taught (mean eval.: 4.8/5.0)	Semester	Enrollment
Fluid Mechanics	F-23	51
	F-22	48
	F-21	30
	F-20	48
Fluid Mechanics Lab	F-23	27
	F-22	35
	F-21	22
	F-20	34
Open Channel Flow	F-26	29
	S-22	11
	S-21	11
	S-20	4
Mechanics of Sediment Transport	S-19	6
	F-22	3
	F-20	3
	F-18	7
Watershed Erosion and Sedimentation	F-25	12
	F-23	7
	F-21	9
	F-19	5
Machine Learning in Water Resources	S-26	9
	S-25	26

SERVICE & PROFESSIONAL ACTIVITIES

Journal Editor

Hydrology and Earth System Sciences (HESS)
Associate Editor (2026 – Present)

Peer Reviewer

Nature Sustainability, Nature Water, Water Research, Water Resources Research, Journal of Hydrology, Hydrology and Earth System Sciences, Hydrological Processes, Journal of Hydraulic Engineering, Journal of Hydrologic Engineering, Geophysical Research Letters, Journal of Geophysical Research: Biogeosciences, Global Biogeochemical Cycles, Freshwater Science, Environmental Earth Sciences, Agriculture Ecosystems and Environment, and the Transactions of the ASABE

Proposal Reviewer

U.S. National Science Foundation (NSF, ad-hoc, panel), German Research Foundation (DFG, ad-hoc), French National Research Agency (ANR, ad-hoc)

University Governance

University level

VT Library Committee
Engineering Representative (2025 – 2028)

College level

KU COE Library Subcommittee
Member (2018 – 2020)
KU COE DEI Subcommittee
Member (2022 – 2024)

Department level

VT EWR Graduate Admissions Committee
Member (2025 – Present)
KU CEAE DEIB Subcommittee
Chair (2020 – 2023)

Professional Committees

Virginia Department of Environmental Quality (VADEQ) Academic Research Group
Member (2026 –Present)
American Geophysical Union (AGU) Water Quality Committee
Member (2019 – Present)
Chair (2021 – 2023)
Deputy Chair (2020 – 2021)
American Society of Civil Engineers (ASCE) Sedimentation Committee
Member (2018 – Present)

Workshop Development

STREAM (Sensor-based Time-series Repository for Environmental Assessment and Modeling) Workshop (2026, February 23–27). University of Arkansas. Fayetteville, Arkansas, USA. 28 workshop participants.

XAI for Catchment Science (2024, September 23–27). Federal University of Mato Grosso do Sul. Campo Grande, Mato Grosso do Sul, Brazil. 13 workshop participants.

Working Groups

Mentoring Institute for Sediment Transport (MIST) Researchers, 2023 – 2024. Funded by the National Science Foundation. Led by Jennifer Duan at the University of Arizona.

John Wesley Powell Center for Analysis and Synthesis, Headwater Modeling Research Working Group, 2023 – 2025. Funded and led by the Environmental Protection Agency and the United States Geological Survey.

International Nitrogen Network (iN-Net), Data and Indicators Working Group, 2025 – 2028. Funded by the National Science Foundation. Led by New York University.

SOCIETY MEMBERSHIPS

Member of the American Society of Civil Engineers (ASCE)
Member of the American Geophysical Union (AGU)

Member of the Association for the Sciences of Limnology and Oceanography (ASLO)
Member of the American Water Works Association (AWWA)
Member of the Water Environment Federation (WEF)
Member of the Tau Beta Pi Engineering Honor Society (TBP)
Member of the Chi Epsilon Civil Engineering Honor Society (XE)
Member of the Omicron Delta Kappa Leadership Society (ODK)

HONORS & AWARDS

NSF CAREER Award (2024)
NSF EPSCoR Track-4 Research Fellow (2023)
KU School of Engineering Bellows Faculty Scholar Award (2023)
KU School of Engineering Miller Faculty Scholar Award (2021)
KU Center for Teaching Excellence Course Transformation Grant (2021)
KU Nominated for Undergraduate Research Mentor Award (2021)
KU One University Open Access Fund Award (2021)
NSF NEON Early Career Scholar (2020)
ASCE ExCEED Teaching Fellow (2019)
Outstanding Reviewer Award, *Journal of Hydrology* (2018)
National Ground Water Association Past President's Award (2017)
Burton E. Heard Graduate Fellowship (2017)
KY-TN American Water Works Association Fellowship (2017)
Burton E. Heard Graduate Fellowship (2016)
Great Lakes National Scholarship (2016)
USGS KY Water Resources Research Institute Research Enhancement (2015)
ASCE J. Waldo Smith Hydraulic Fellowship (2015)
Robert Eugene Fish Hydraulic Fellowship (2014)